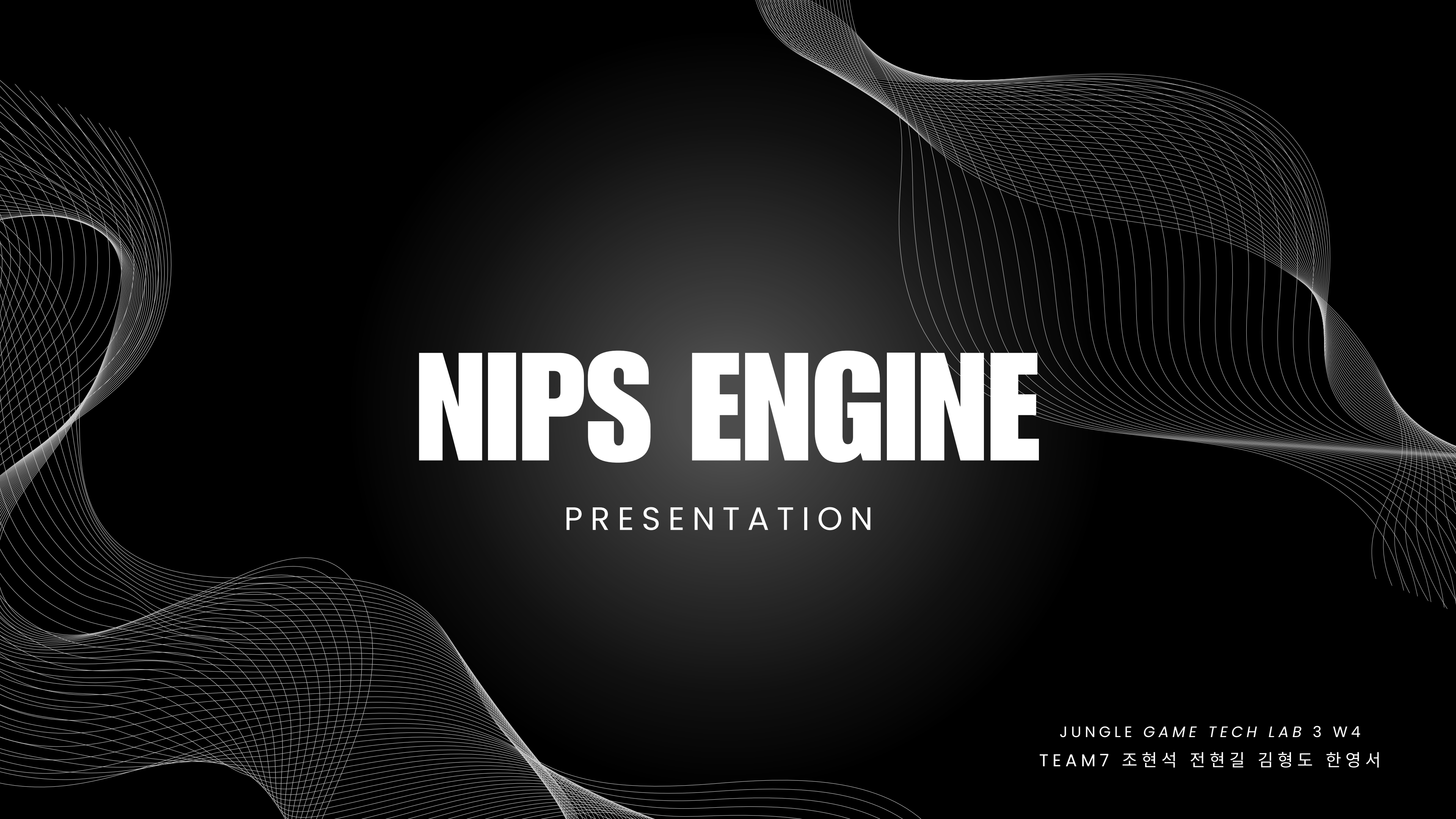




NIPS ENGINE

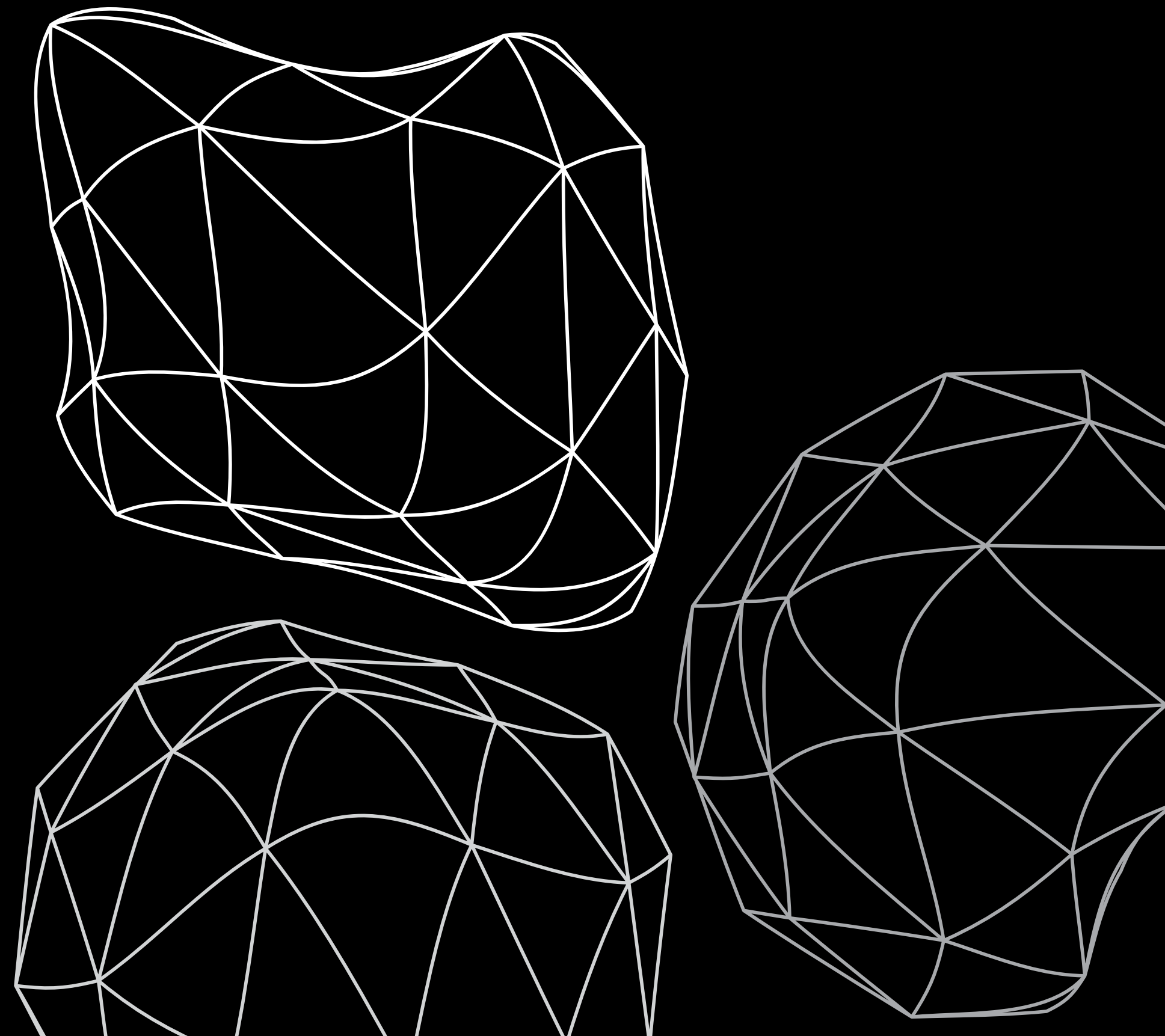
PRESENTATION

JUNGLE GAME TECH LAB 3 W4
TEAM7 조현석 전현길 김형도 한영서

최적화 기법

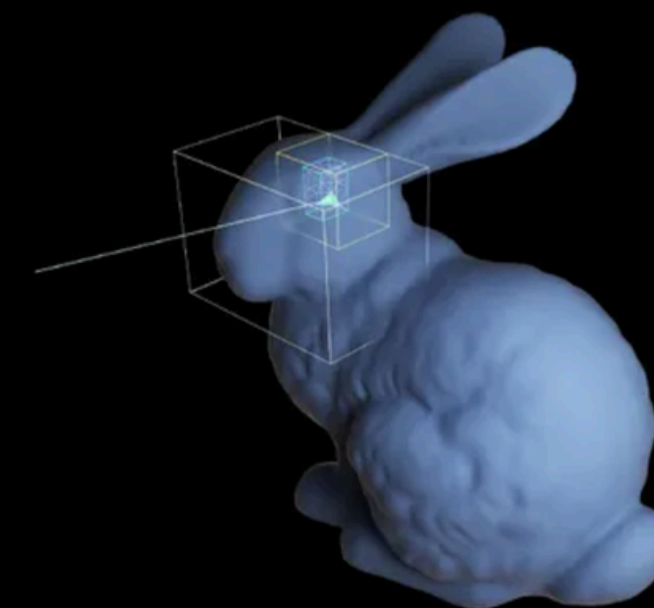
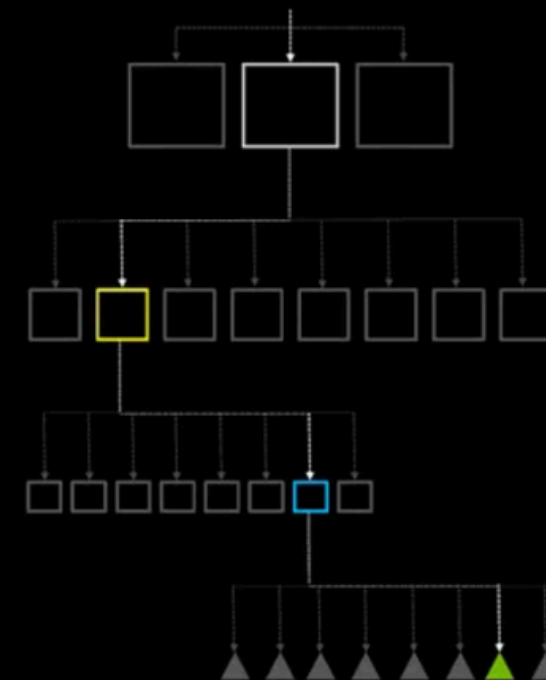
- ✓ BVH (TLAS – BLAS 구조)
- ✓ Frustum Culling
- ✓ Occlusion Culling
- ✓ LOD
- ✓ CPU 친화적 자료구조
- ✓ SIMD

GAME TECH LAB

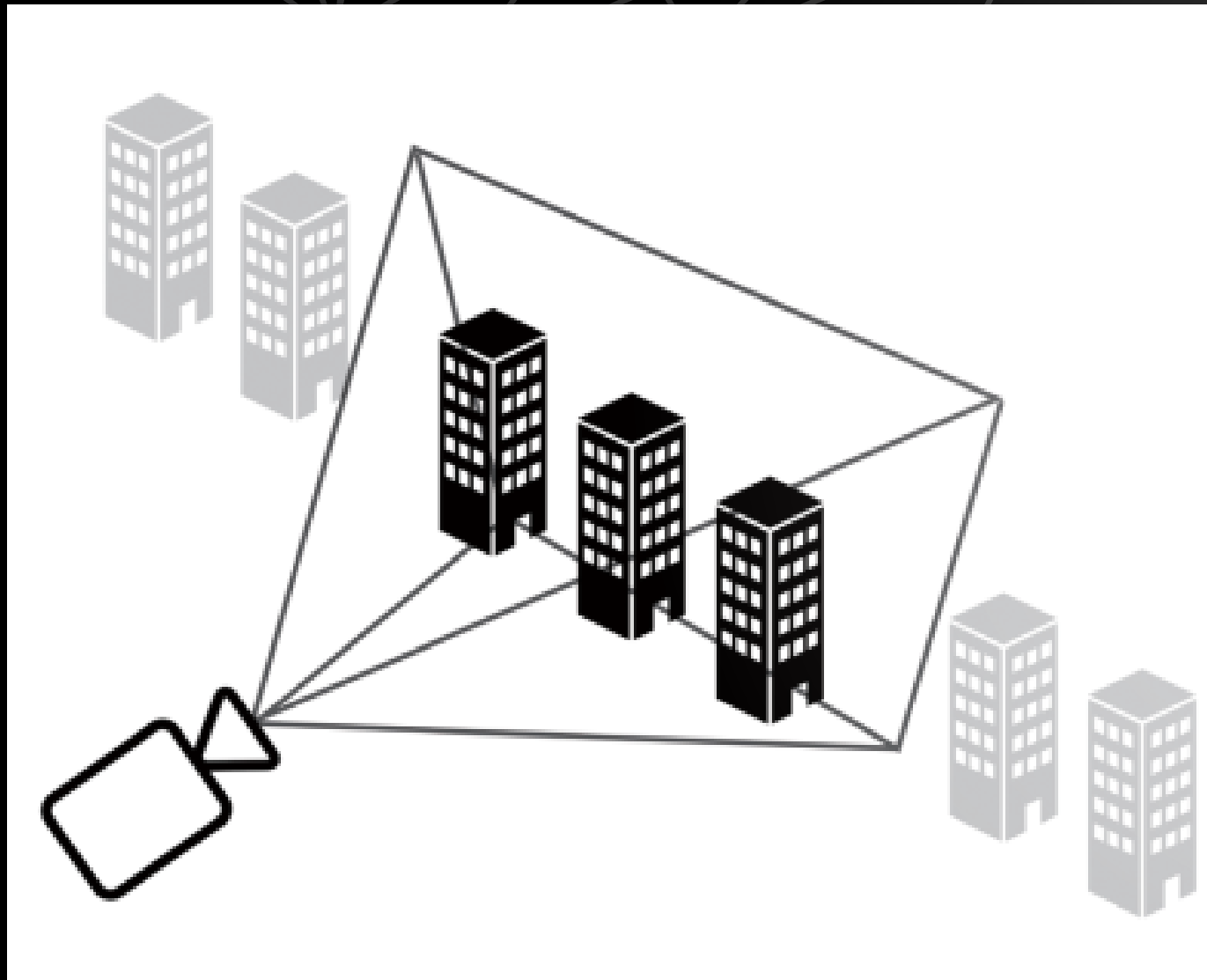




Massive Improvement in Search Efficiency



Frustum Culling



AVG: 65.2 FPS (15.34 ms)
MAX: 17.18 ms

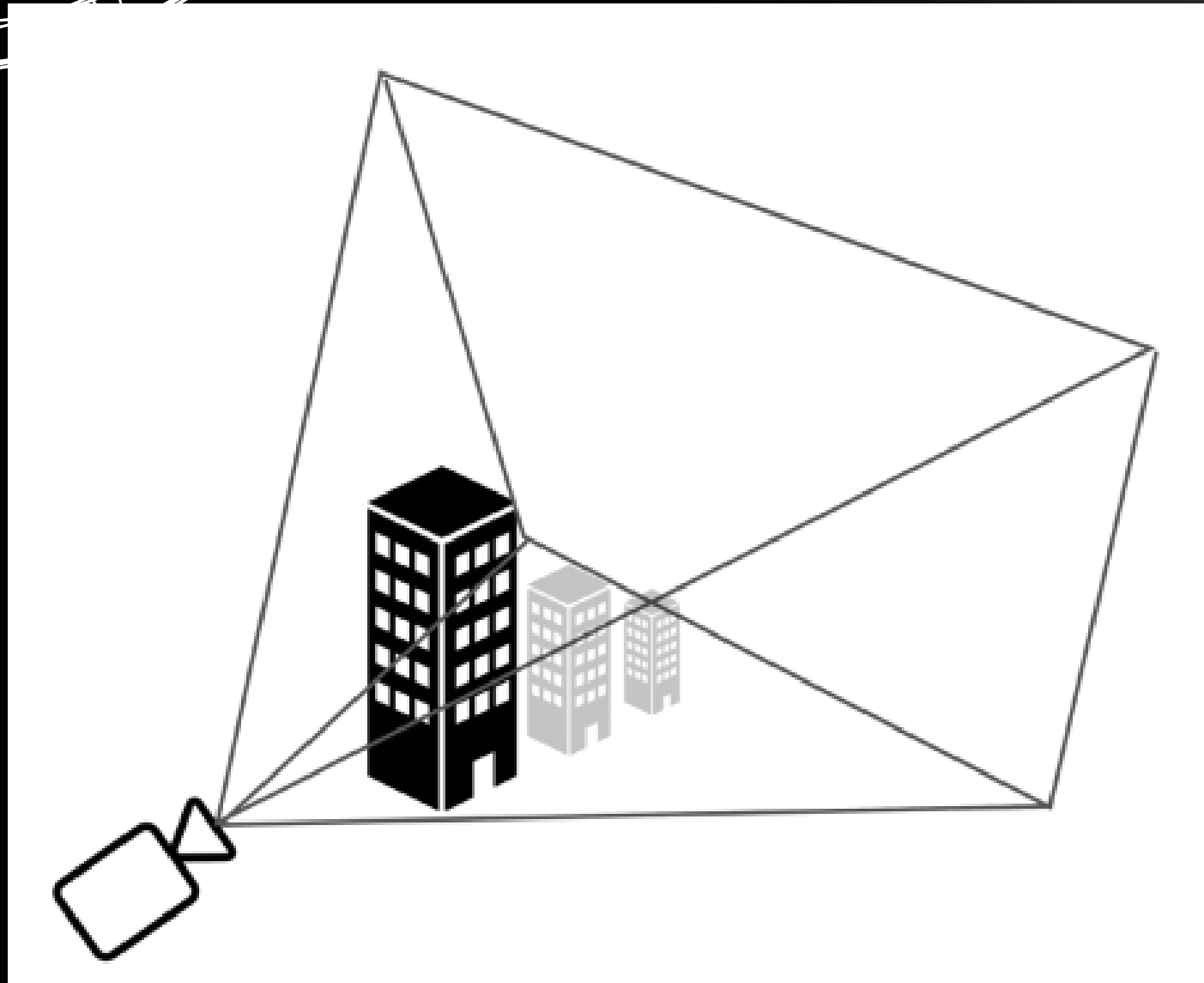


AVG: 146.9 FPS (6.81 ms)
MAX: 10.25 ms

DefaultScene 기준



Occlusion Culling



x 2.3

AVG: 80.2 FPS (12.47 ms)
MAX: 15.86 ms



AVG: 184.8 FPS (5.41 ms)
MAX: 7.24 ms

LEVEL OF DETAIL



LOD 0



LOD 1



LOD 2



LOD 3



LOD 4

화면에 차지하는 비율을 구해서 LOD 적용
UV, NORMAL은 같지만 위치가 같은 중복 정점은 해시맵으로 관리

GPU 데이터 전송

Vertex Buffer

- 16bytes alignment
- 위치 12byte + UV 4byte 압축

Structured Buffer

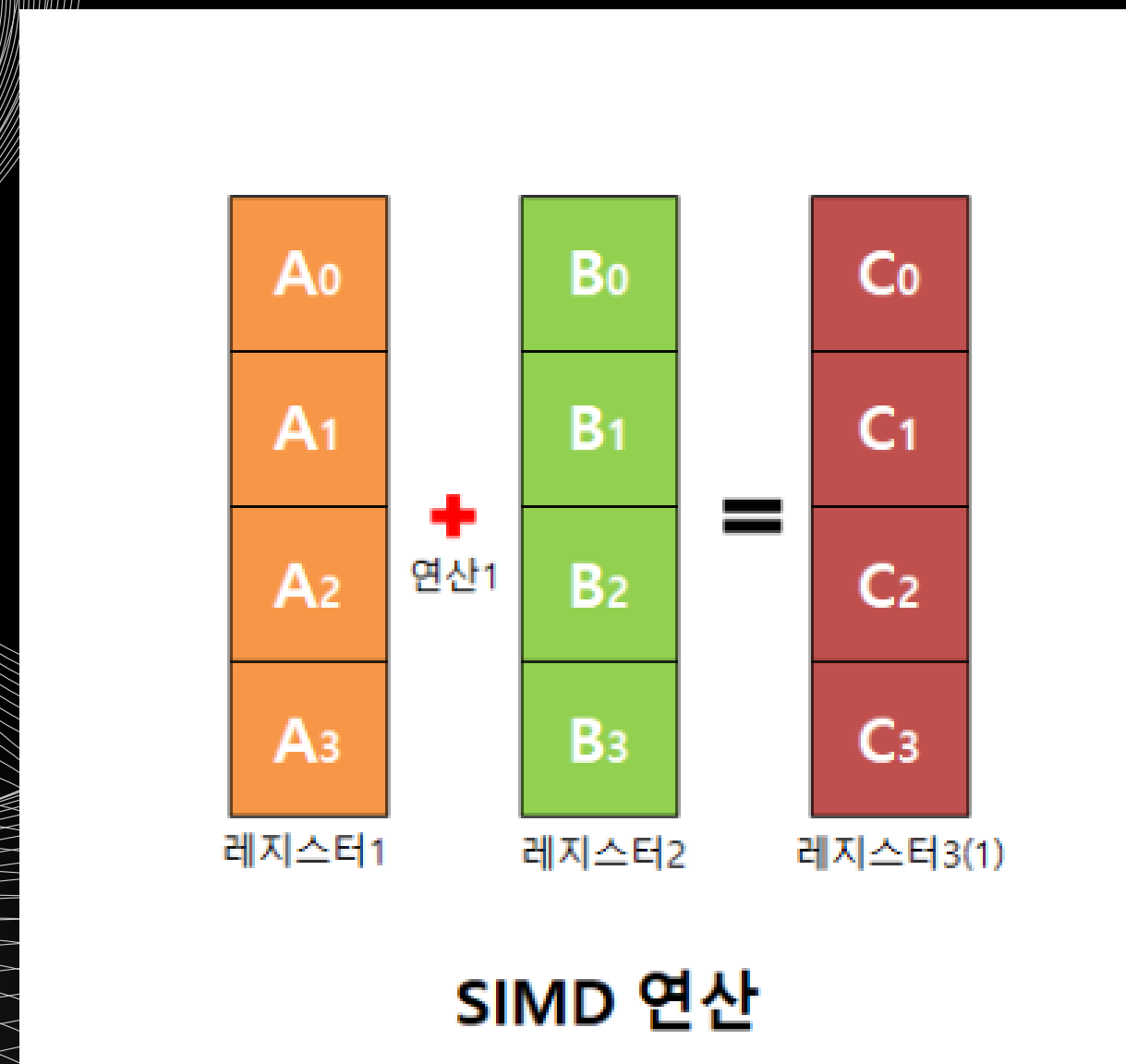
- 모든 Model 행렬 미리 업로드

Constant Buffer

- VP 행렬 곱한 뒤 전송
- Diffuse Color/Map 저장

SIMD

Simple Instruction Multiple Data



CPU 멀티코어를 이용한 병렬연산

TRIES..

Texture 거리별 Mip-Map 적용

KD-Tree

메시, 머터리얼, 텍스처 기준 컴포넌트 정렬

GAME TECH LAB

